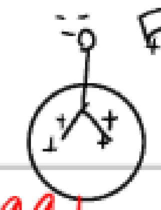
Electric Charge & Electric Field

1. All charge is quantized. Algebraic sum of all the electric charges in any close system is constant.

2. conductor: silver, gold, copper, steel, sea water
   insulator: rubber, glass, oil, diamond, dry wood
   semiconductor

3. plastic rods, after being rubbed with fur, repel each other (−)
   glass rods, - - - silk, repel each other (+)
   Both can attract the rod it rubbed

4. charging by induction



5. Coulomb's Law $F=\frac{1}{4\pi\varepsilon_0}\frac{|q_1q_2|}{r^2}$

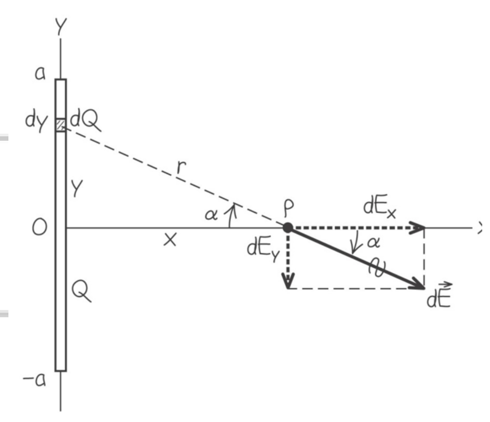
$\varepsilon_0=8.854\times10^{-12}\,C^2/N\cdot m^2$ $\quad \frac{1}{4\pi\varepsilon_0}=k=8.988\times10^{9}\,N\cdot m^2/c^2$

6. Electric field $\vec{E}=\frac{\vec{F_0}}{q_0}$

$\vec{E}=\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$ (point charge)

7. 不同类型电场积分

①



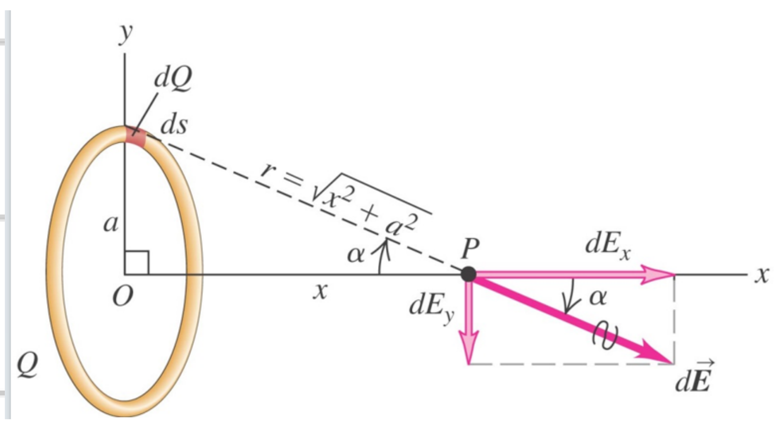
$E_x=\frac{1}{4\pi\varepsilon_0}\frac{Qx}{2a}\int_{-a}^{a}\frac{dy}{(x^2+y^2)^{\frac{3}{2}}}=\frac{Q}{4\pi\varepsilon_0}\frac{1}{x\sqrt{x^2+a^2}}$

$E_y=0$

$\vec{E}=\frac{Q}{4\pi\varepsilon_0}\frac{1}{x\sqrt{x^2+a^2}}\hat{i}$

(可推导得到 高斯定理 证明出的 $\vec{E}=\frac{\lambda}{2\pi\varepsilon_0 x}$，对于长直导线)

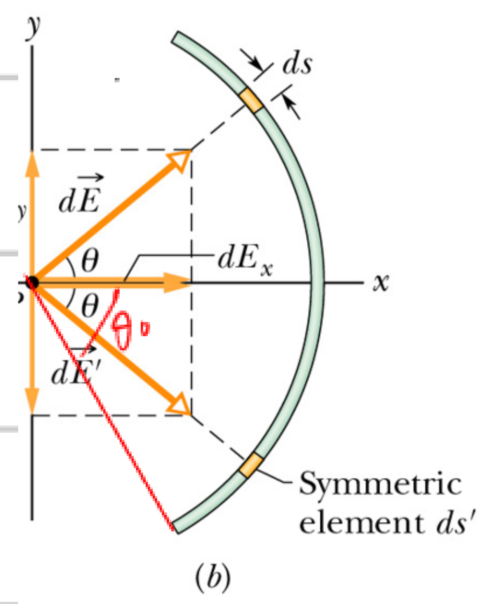
②



$dE_x=\frac{1}{4\pi\varepsilon_0}\frac{x\,dQ}{(x^2+a^2)^{\frac{3}{2}}}$

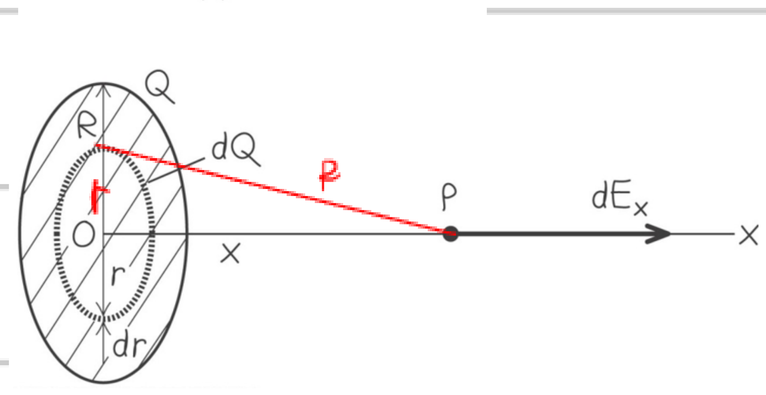
$\vec{E}=E_x\hat{i}=\frac{1}{4\pi\varepsilon_0}\frac{Qx}{(x^2+a^2)^{\frac{3}{2}}}\hat{i}$ (没用积分)

③



$E_x=\cos\theta\frac{kdq}{R^2}=\cos\theta\frac{k\lambda ds}{R^2}=\frac{k\lambda}{R}\int_{-\theta_0}^{\theta_0}\cos\theta\,d\theta=\frac{2k\lambda}{R}\sin\theta_0$

④



$dE_x=\frac{1}{4\pi\varepsilon_0}\frac{dQ}{R^2}=\frac{1}{4\pi\varepsilon_0}\frac{(2\pi\sigma r\,dr)x}{(x^2+r^2)^{\frac{3}{2}}}$

$E_x=\frac{\sigma x}{2\varepsilon_0}\int_0^R\frac{r\,dr}{(x^2+r^2)^{\frac{3}{2}}}$

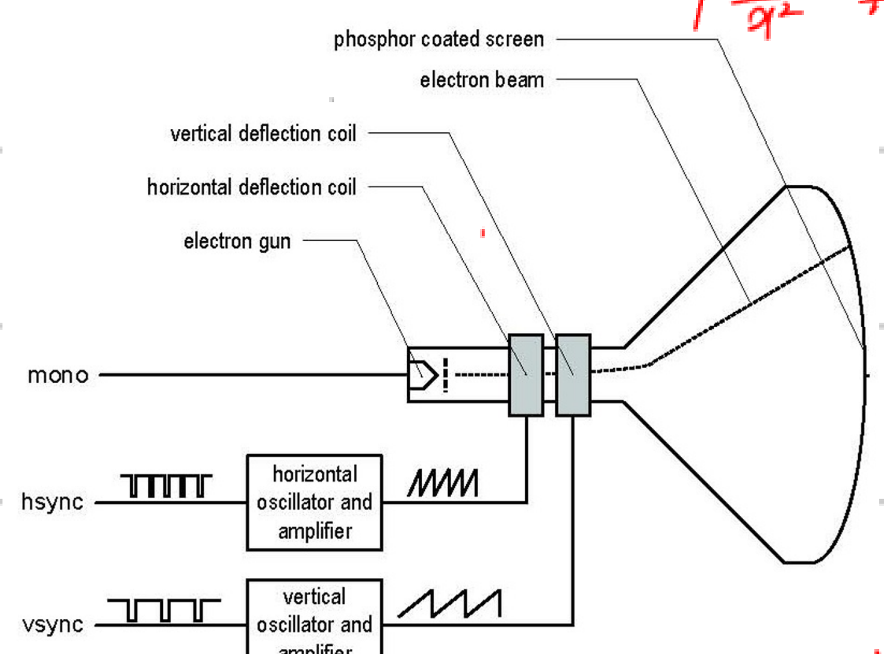
$=\frac{\sigma x}{2\varepsilon_0}\left[-\frac{1}{\sqrt{x^2+R^2}}+\frac{1}{x}\right]$

$=\frac{\sigma}{2\varepsilon_0}\left[1-\frac{1}{\sqrt{\frac{R^2}{x^2}+1}}\right]$

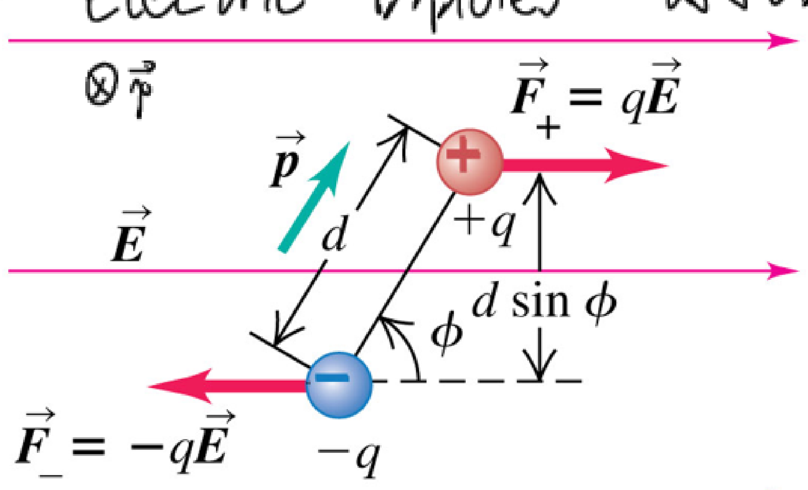
当 $R\gg x$ $\quad \boxed{E_x=\frac{\sigma}{2\varepsilon_0}}$

⑤ 球壳内 $\vec{E}$ 均为 0

8. 阴极射线管



9. Electric Dipoles — 一对等电量但异号电荷



$|p|=qd$ 电偶极矩 (dipole moment) $\vec{p}$ 从负电荷指向正电荷

$\tau=pE\sin\phi$

$\vec{\tau}=\vec{p}\times\vec{E}$ (torque on a dipole moment)

Gauss's Law

1. Electric Flux $\Phi=\vec{E}\cdot\vec{A}$ $\quad \Phi_E=EA\cos\phi$

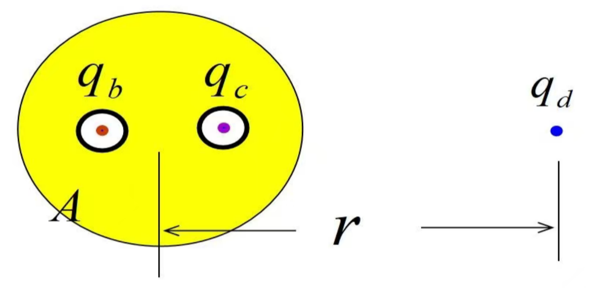
2. Gauss's Law $\Phi_E=\oint\vec{E}\cdot d\vec{A}=\frac{Q_{encl}}{\varepsilon_0}$

3. 电场强度的散度 $\nabla\cdot E=\frac{\rho(r)}{\varepsilon_0}$ 用高斯定律 & 库仑定律推导 ρ: charge density

4. 导体材料任意一点场强为 0；但不针对内部空心

5. 电场屏蔽：

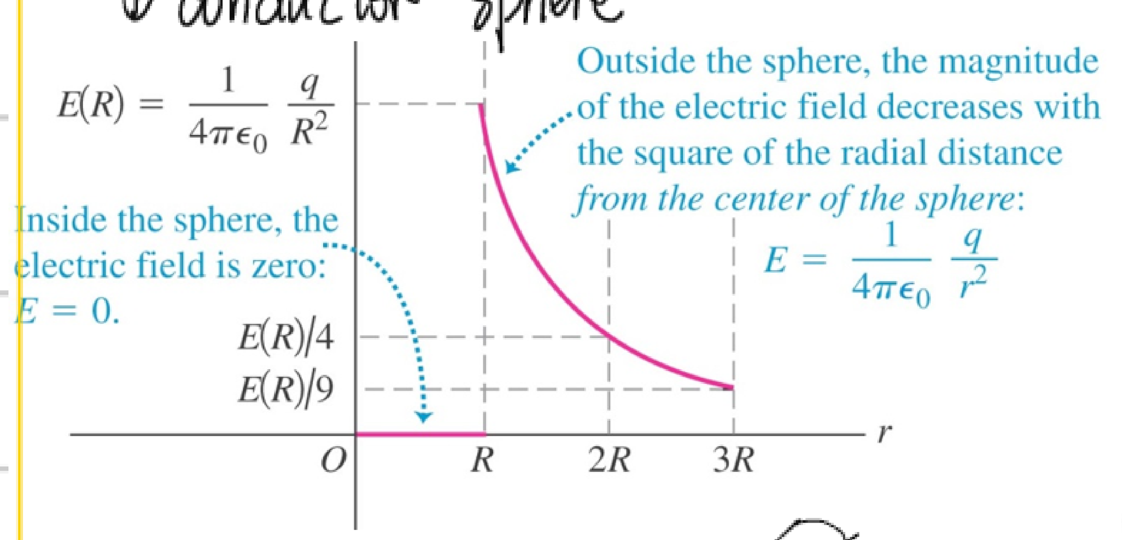
腔内的电场仅由腔内 q 和腔内表面的感应电荷 $(-q)$ 共同决定，与腔内电荷位置、腔体的几何结构有关。



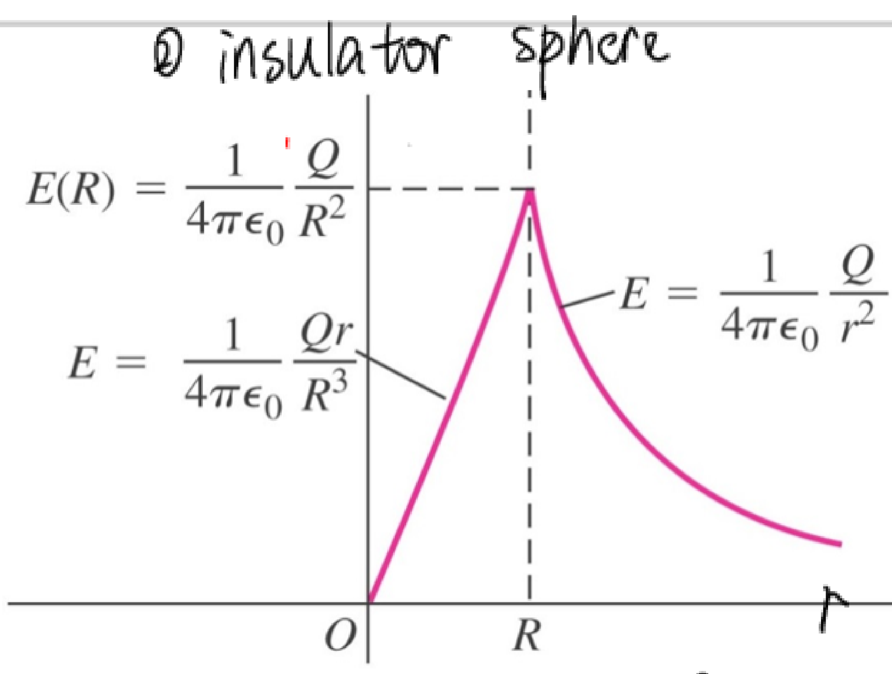
解: 两空腔内的电场都不受外界影响;内表面感应电荷均匀分布，故它们在腔内产生的场强处处为零，q_b、q_c 受力为零。

6. 场强分布

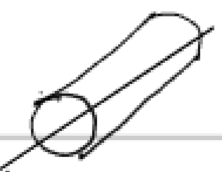
① conductor sphere



② insulator sphere



③ line charge

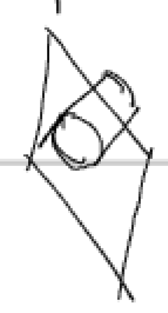


$E\cdot2\pi r\,l=\frac{\lambda l}{\varepsilon_0}\Rightarrow E=\frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$

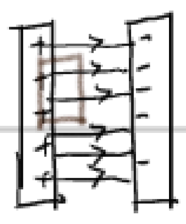
④ sheet of charge (infinite)

$2EA=\frac{\sigma A}{\varepsilon_0}$

$\Rightarrow\boxed{E=\frac{\sigma}{2\varepsilon_0}}$



⑤ 平行电容板



$E=\frac{\sigma}{\varepsilon_0}$

Electric Potential

1. $W_{a\to b}=U_a-U_b=-(U_b-U_a)=-\Delta U$

$=\int_a^b\vec{F}\cdot d\vec{l}$

2. $U=\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r}$ electrical potential energy，正负号由 q、q_0 正负性决定

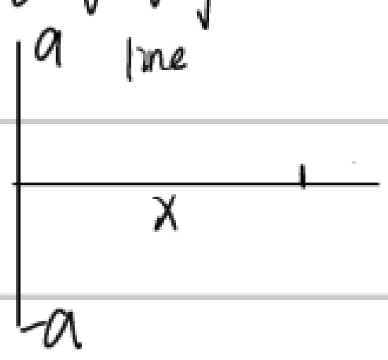
标量叠加 $U=\frac{q_0}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1}+\frac{q_2}{r_2}+\frac{q_3}{r_3}\cdots\right)$

3. electric potential: $V=\frac{U}{q_0}=\frac{1}{4\pi\varepsilon_0}\frac{q}{r}$

$V_{total}=\frac{1}{4\pi\varepsilon_0}\sum\frac{q_i}{r_i}=\frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r}$

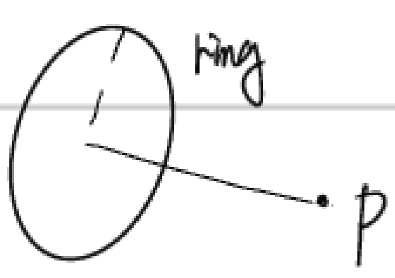
4. 不同电势分布

① line



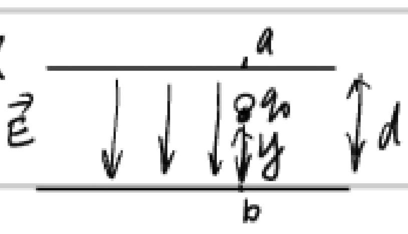
$V=\frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\int_{-a}^{a}\frac{dy}{\sqrt{x^2+y^2}}=\frac{1}{4\pi\varepsilon_0}\frac{Q}{2a}\ln\left(\frac{\sqrt{a^2+x^2}+a}{\sqrt{a^2+x^2}-a}\right)$

② ring



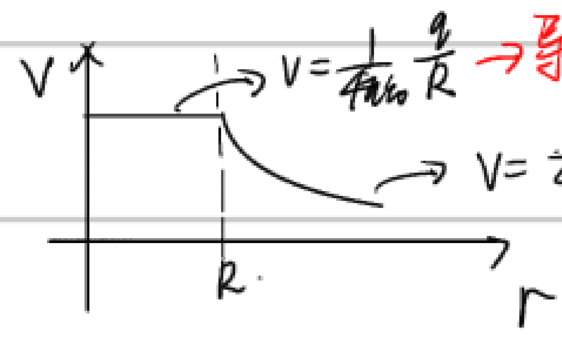
$V=\frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r}=\frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{x^2+a^2}}$

③ 平行电容板

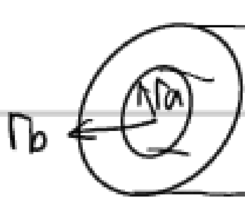


$V(y)=\frac{U(y)}{q_0}=Ey$

④ conductor sphere



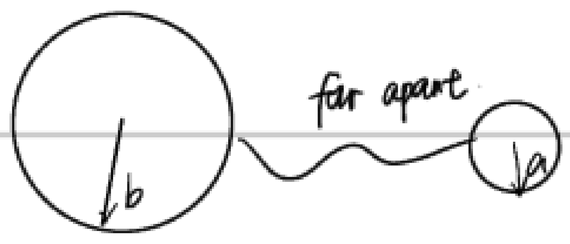
⑤ 柱形电容器



$E=\frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$ $\quad V_a-V_b=\int_a^b\vec{E}\cdot d\vec{l}=\frac{\lambda}{2\pi\varepsilon_0}\int_{r_a}^{r_b}\frac{dr}{r}=\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_b}{r_a}$

$C=\frac{Q}{V_{ab}}=\frac{\lambda L}{\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_b}{r_a}}=\frac{2\pi\varepsilon_0 L}{\ln(r_b/r_a)}$

$\frac{C}{L}=\frac{2\pi\varepsilon_0}{\ln(\frac{r_b}{r_a})}$

5. 导线相连构造等势体 

$q_a+q_b=q$

$V_b=\frac{q_b}{4\pi\varepsilon_0 b}=V_a=\frac{q_a}{4\pi\varepsilon_0 a}$

$q_b=q\frac{b}{a+b}\quad q_a=q\frac{a}{a+b}$

if $b\gg a \Rightarrow E_a\gg E_b$

6. 立体角证明带电球壳内部电场为0

$$dE=\frac{\rho_s}{4\pi\epsilon_0}\left(\frac{ds_1}{r_1^2}-\frac{ds_2}{r_2^2}\right).\qquad \Omega=\frac{S}{r^2}$$

ngle $d\Omega$ equals

$$d\Omega=\frac{ds_1}{r_1^2}\cos\alpha=\frac{ds_2}{r_2^2}\cos\alpha.$$

expressions of dE and $d\Omega$, we find that

$$dE=\frac{\rho_s}{4\pi\epsilon_0}\left(\frac{d\Omega}{\cos\alpha}-\frac{d\Omega}{\cos\alpha}\right)=0.$$

7. potential gradient $\vec{E}=-\vec{\nabla}V$

$\vec{E}=-\left(\hat{i}\frac{\partial V}{\partial x}+\hat{j}\frac{\partial V}{\partial y}+\hat{k}\frac{\partial V}{\partial z}\right)$

球坐标系中 $\nabla f(r,\theta,\phi)=\frac{\partial f}{\partial r}\hat{r}+\frac{1}{r}\frac{\partial f}{\partial\theta}\hat{\theta}+\frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}\hat{\phi}$

Capacitance and Dielectrics

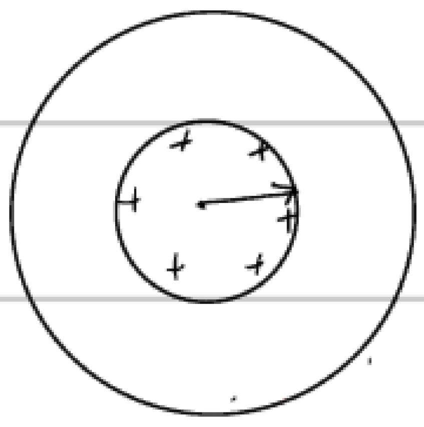
1. Any two conductors separated by an insulator form a capacitor. $C=\frac{Q}{V_{ab}}$

2. 不同的电容

①平行板电容 $E=\frac{\sigma}{\varepsilon_0}=\frac{Q}{\varepsilon_0 A}\quad V_{ab}=Ed=\frac{Qd}{\varepsilon_0 A}$

$C=\frac{Q}{V}=\frac{\varepsilon_0 A}{d}$

②球形电容器：



$V_{ab}=V_a-V_b=\frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_a}-\frac{1}{r_b}\right)$

$C=\frac{Q}{V_{ab}}=4\pi\varepsilon_0\frac{r_ar_b}{r_b-r_a}$

③柱形电容器 见上页最末

3. 电容器串并联 串联 $\frac{1}{C_{eq}}=\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\cdots$

并联 $C_{eq}=C_1+C_2+C_3+\cdots$

4. 电容器的 potential energy $U=\frac{Q^2}{2C}=\frac{1}{2}CV^2=\frac{1}{2}QV$

energy density $u=\frac{1}{2}\varepsilon_0E^2$

5. Dielectrics 介电材料 dielectric constant $k=\frac{C}{C_0}>1$

介电材料的加入会相对削弱平行板间的场强、电势差

$E=\frac{E_0}{k}$, $E_0=\frac{\sigma}{\varepsilon_0}$, $E=\frac{\sigma-\sigma_i}{\varepsilon_0}$ $\Rightarrow \sigma_i=\sigma\left(1-\frac{1}{k}\right)$ induced surface charge density

$\varepsilon=k\varepsilon_0$

$E=\frac{\sigma}{\varepsilon}\quad C=kC_0=k\frac{\varepsilon_0A}{d}=\frac{\varepsilon A}{d}$ (density) $u=\frac{1}{2}k\varepsilon_0E^2=\frac{1}{2}\varepsilon E^2$

总结：$\varepsilon\uparrow_k\quad C\uparrow_k\quad u\uparrow_k\quad E\downarrow_k$.

6. dielectric breakdown

If the electric field is strong enough, *dielectric breakdown* occurs and the dielectric becomes a conductor.

The *dielectric strength* is the maximum electric field the material can withstand before breakdown occurs.

7. dielectric 介入时 Gauss's Law

$\oint k\vec{E}\cdot d\vec{A}=\frac{Q_{encl-free}}{\varepsilon_0}\qquad \oint\vec{E}\cdot d\vec{A}=\frac{Q_{encl-free}}{\varepsilon}$

Current, Resistance, and Electromotive Force

1. $I=\frac{dQ}{dt}=nqv_dA$ (τ: mean free time)

电流密度 $J=\frac{I}{A}=nqv_d\qquad v_d=\frac{q\tau}{m}\vec{E}$

$=\frac{nq^2\tau}{m}\vec{E}$

resistivity 电阻率 $\rho=\frac{E}{J}=\frac{m}{nq^2\tau}$ (只与材料、温度有关)

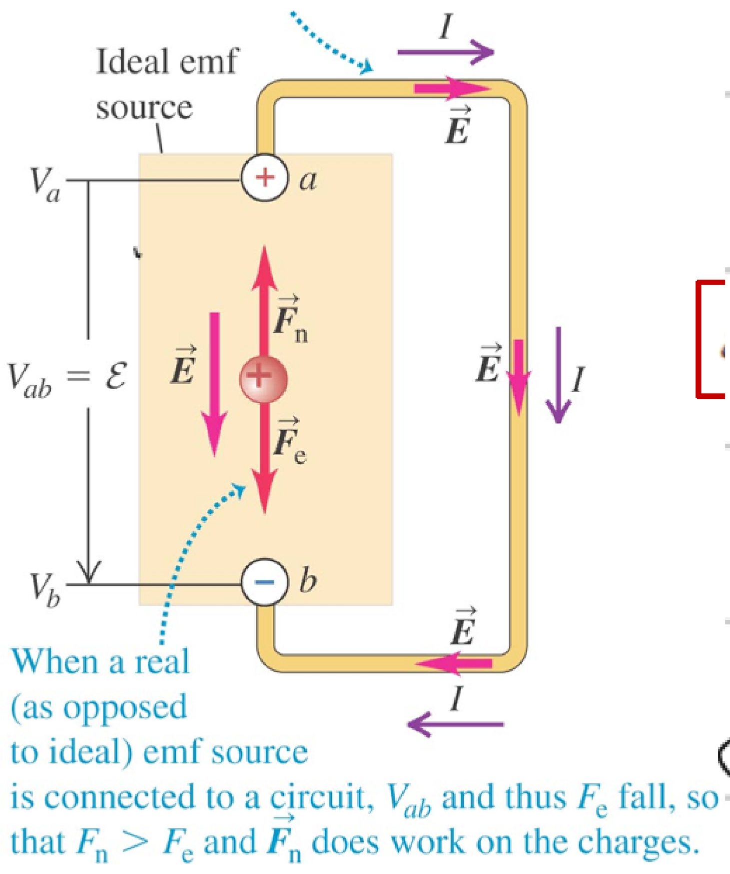
$\rho(T)=\rho_0[1+\alpha(T-T_0)]$

电导率 $=\frac{1}{\rho}$ conductivity

resistance：电阻 $R=\rho\frac{L}{A}\quad V=IR$ (Ohm's Law)

2. 密合回路或完整回路中才有稳定电流

3. EMF：electromotive force, not real force

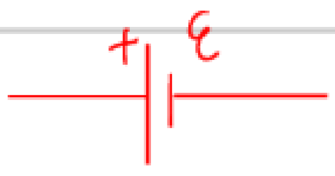


$\mathcal{E}=V_{ab}=IR$ (ideal source of emf)

$V_{ab}=\mathcal{E}-Ir$ (terminal voltage)

$\mathcal{E}I-I^2r_{inner}$ (power output)

When the emf source is not part of a closed circuit, $F_n=F_e$ and there is no net motion of charge between the terminals.

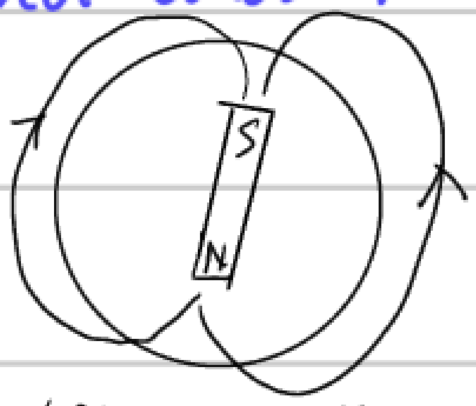

长的一般是正极

4. Hall Effect

$qE=qvB\qquad J=nqv_d\qquad nq=\frac{-JB}{E}$

Magnetic Field and Magnetic Forces

1. 地球北极是南磁极



2. A moving charge generates a magnetic field that depends on the velocity of the charge

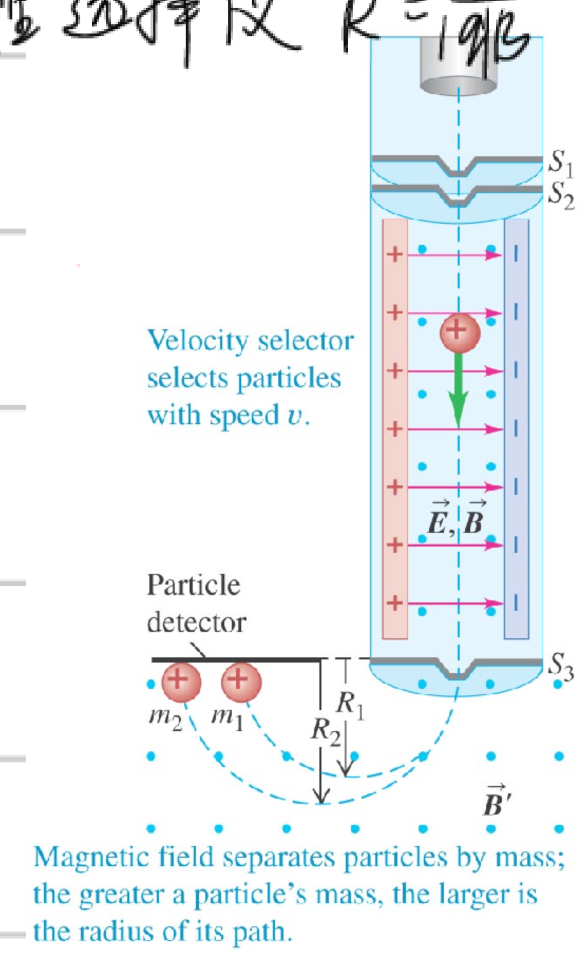
3. Magnetic flux through any closed surface is 0. No magnetic monopole. Magnetic field lines have no ends

4. magnetic force $\vec{F}=q\vec{v}\times\vec{B}$

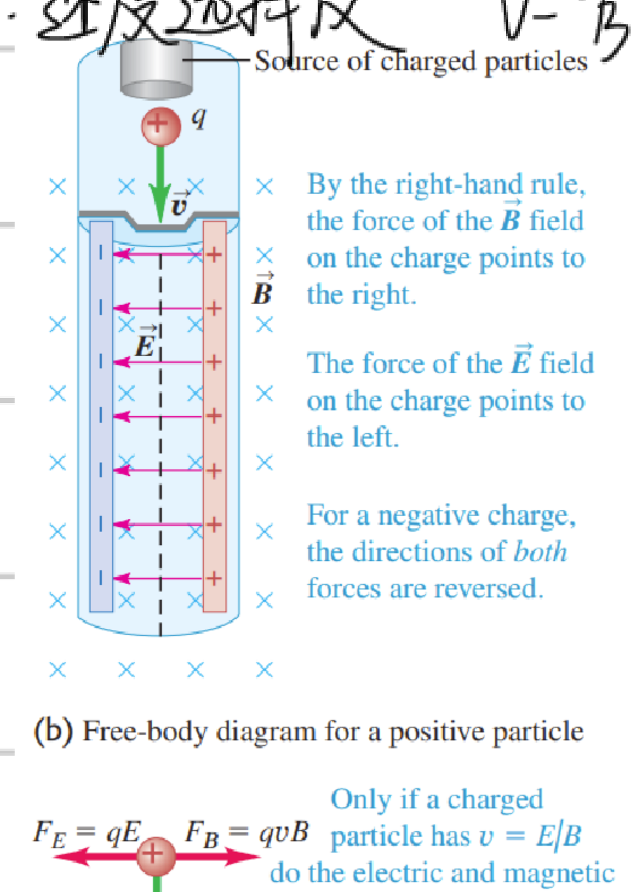
5. cyclotron frequency $f_c=\frac{qB}{2\pi m}$ 回旋频率

角频率 $\omega_c=\frac{qB}{m}$

6. 质量选择仪 $R=\frac{mv}{|q|B}$



7. 速度选择仪 $V=\frac{E}{B}$



(b) Free-body diagram for a positive particle

$F_E=qE\quad F_B=qvB$ Only if a charged particle has $v=E/B$ do the electric and magnetic forces cancel. All other particles are deflected.

7. 一段通电导线所受洛伦兹力 $F=ILB$ $\vec F = I\vec l\times\vec B$

8. 法拉第电磁感应定律 $\mathcal{E} = -\frac{d\Phi}{dt}$ $\mathcal{E}=BLv$

9. magnetic moment: $\mu = IA \rightarrow \mu = NIA$

torque: $\vec\tau = \vec\mu\times\vec B$

Sources of Magnetic Field

1. Bohr Magneton $\mu = IA = I(\pi r^2) = \frac{ev}{2\pi r}(\pi r^2) = \frac{evr}{2}$

$T=\frac{2\pi r}{v}$ $I=\frac{e}{T}=\frac{ev}{2\pi r}$

又由于电子角动量 $L=mvr=\frac{h}{2\pi}$ $\mu=\frac{e}{2m}L$

$= \frac{e}{2m}\frac{h}{2\pi}$

2. 不同来源的磁场

① a point charge moving with constant velocity

$\vec B = \frac{\mu_0}{4\pi}\frac{q\vec v\times\hat r}{r^2}$ 注：A移动产生的磁场作用在B上，那么 V 指 V_A

② 小段导线 $\boxed{d\vec B = \frac{\mu_0}{4\pi}\frac{Id\vec l\times\hat r}{r^2}}$ （参照点距离相较导线长度很大时可用）

③ 长直导线 $\boxed{B=\frac{\mu_0 I}{2\pi r}}$

两根平行长直导线的相互作用力 $\boxed{F=I'LB=\frac{\mu_0 II'L}{2\pi r}}$

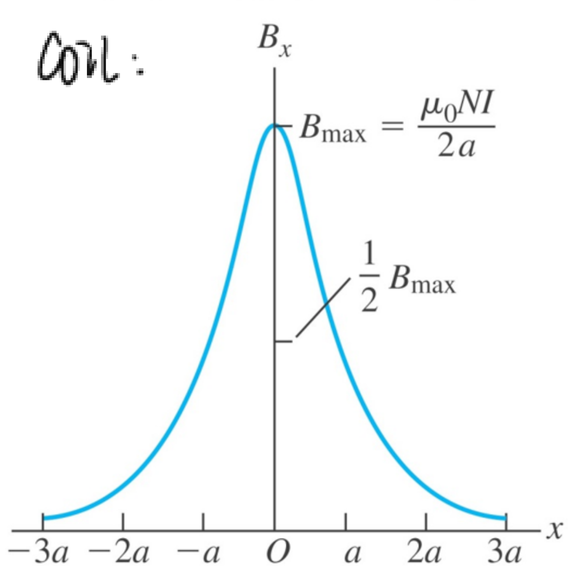
$\frac{F}{L}=\frac{\mu_0 II'}{2\pi r}$

同向电流相互吸引，异向电流相互排斥

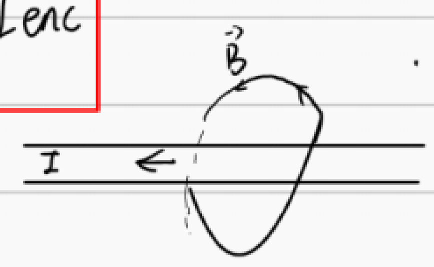
④ 环形导线对轴线上一点 $\boxed{B_x = \frac{\mu_0 I a^2}{2(x^2+a^2)^{\frac{3}{2}}}}$ ← radius of loop

多层环形导线 $B_x = \frac{\mu_0 I a^2}{2(x^2+a^2)^{\frac{3}{2}}}$ $x=0$ $B_{x\max} = \frac{\mu_0 NI}{2a}$

Coil:



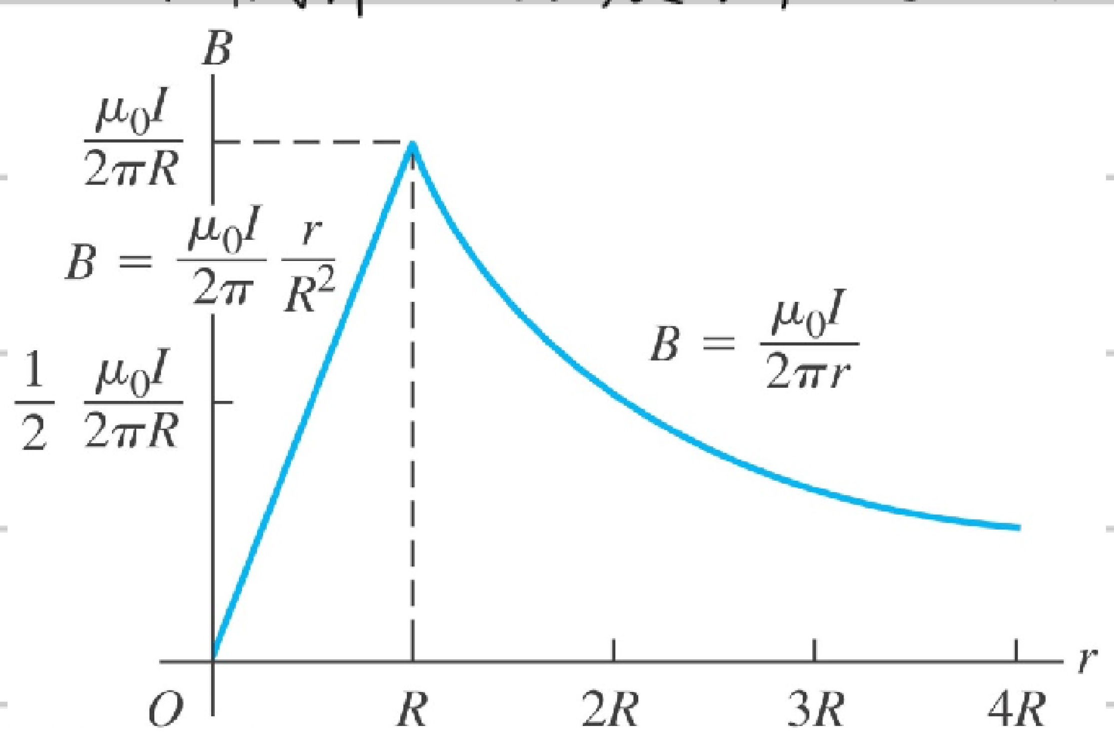
3. 安培定律 $\boxed{\oint_C \vec B\cdot d\vec l = \mu_0 I_{enc}}$



理想螺线管 { 内部磁场均匀 / 外部磁场为0

*避坑 算相互作用力时，同时考虑电场和磁场

半径为R的柱形导体磁场强度分布（安培定律）



积分表

① $\int \frac{1}{(x^2+y^2)^{\frac{3}{2}}} dy = \frac{y}{x^2\sqrt{x^2+y^2}} + C$

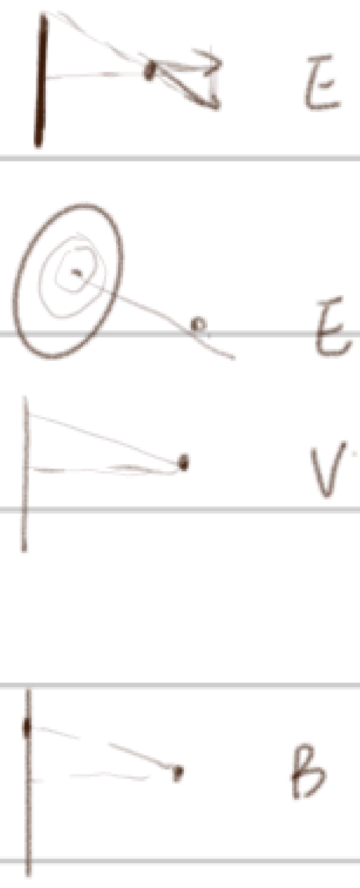
② $\int \frac{y}{(x^2+y^2)^{\frac{3}{2}}} dy = -\frac{1}{\sqrt{x^2+y^2}} + C$

③ $\int \frac{1}{(x^2+y^2)^{\frac{1}{2}}} dy = \ln(\sqrt{x^2+y^2} + y)$

④ $\int \frac{x}{(x^2+y^2)^{\frac{3}{2}}} dy = \frac{y}{x\sqrt{y^2+x^2}} + C$

⑤ $\int \frac{y^3}{(x^2+y^2)^{\frac{3}{2}}} dy = \frac{y^2+2x^2}{\sqrt{y^2+x^2}} + C$

$\int \frac{y^3}{(x^2-y^2)^{\frac{3}{2}}} dy = \frac{-y^2+2x^2}{\sqrt{-y^2+x^2}} + C$



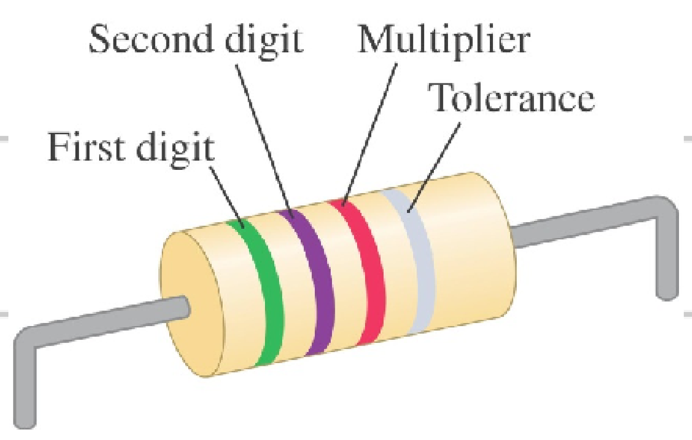
Name	Symbol	Value
Speed of light in vacuum	c	2.99792458×10^8 m/s
Magnitude of charge of electron	e	$1.602176634 \times 10^{-19}$ C
Gravitational constant	G	$6.67408(31) \times 10^{-11}$ $\mathrm{N \cdot m^2/kg^2}$
Planck's constant	h	$6.62607015 \times 10^{-34}$ J · s
Boltzmann constant	k	1.380649×10^{-23} J/K
Avogadro's number	N_A	$6.02214076 \times 10^{23}$ molecules/mol
Gas constant	R	8.314462618 . . . J/mol · K
Mass of electron	m_e	$9.10938356(11) \times 10^{-31}$ kg
Mass of proton	m_p	$1.672621898(21) \times 10^{-27}$ kg
Mass of neutron	m_n	$1.674927471(21) \times 10^{-27}$ kg
Magnetic constant	μ_0	$1.25663706 \times 10^{-6}$ Wb/A · m (approximate) $\cong 4\pi \times 10^{-7}$ Wb/A · m
Electric constant	$\epsilon_0 = 1/\mu_0 c^2$	$8.854187817 \times 10^{-12}$ $\mathrm{C^2/N \cdot m^2}$ (approximate)
	$1/4\pi\epsilon_0$	8.987551787×10^9 $\mathrm{N \cdot m^2/C^2}$ (approximate)

Power of ten	Prefix	Abbreviation	Pronunciation
10^{-24}	yocto-	y	*yoc*-toe
10^{-21}	zepto-	z	*zep*-toe
10^{-18}	atto-	a	*at*-toe
10^{-15}	femto-	f	*fem*-toe
10^{-12}	pico-	p	*pee*-koe
10^{-9}	nano-	n	*nan*-oe
10^{-6}	micro-	μ	*my*-crow
10^{-3}	milli-	m	*mil*-i
10^{-2}	centi-	c	*cen*-ti
10^{3}	kilo-	k	*kil*-oe
10^{6}	mega-	M	*meg*-a
10^{9}	giga-	G	*jig*-a or *gig*-a
10^{12}	tera-	T	*ter*-a
10^{15}	peta-	P	*pet*-a
10^{18}	exa-	E	*ex*-a
10^{21}	zetta-	Z	*zet*-a
10^{24}	yotta-	Y	*yot*-a

angular acceleration	radian per second squared	$\mathrm{rad/s^2}$	
force	newton	N	$\mathrm{kg \cdot m/s^2}$
pressure (mechanical stress)	pascal	Pa	$\mathrm{N/m^2}$
kinematic viscosity	square meter per second	$\mathrm{m^2/s}$	
dynamic viscosity	newton-second per square meter	$\mathrm{N \cdot s/m^2}$	
work, energy, quantity of heat	joule	J	N · m
power	watt	W	J/s
quantity of electricity	coulomb	C	A · s
potential difference, electromotive force	volt	V	J/C, W/A
electric field strength	volt per meter	V/m	N/C
electrical resistance R	ohm	Ω	V/A
capacitance C	farad	F	A · s/V
magnetic flux Φ	weber	Wb	V · s
inductance L	henry	H	V · s/A
magnetic flux density	tesla	T	$\mathrm{Wb/m^2}$
magnetic field strength	ampere per meter	A/m	
magnetomotive force	ampere	A	

Table 25.3 Color Codes for Resistors

Color	Value as Digit	Value as Multiplier
Black	0	1
Brown	1	10
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Gray	8	10^8
White	9	10^9



Integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1) \qquad \int \frac{dx}{x} = \ln x \qquad \int e^{ax}\,dx = \frac{1}{a}e^{ax}$$

$$\int \sin ax\,dx = -\frac{1}{a}\cos ax \qquad \int \cos ax\,dx = \frac{1}{a}\sin ax \qquad \int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a}$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x+\sqrt{x^2+a^2}\right) \qquad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} \qquad \int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{1}{a^2}\frac{x}{\sqrt{x^2+a^2}}$$

$$\int \frac{x\,dx}{(x^2+a^2)^{3/2}} = -\frac{1}{\sqrt{x^2+a^2}}$$

$f(x) = \sum \frac{f^{n}(0)}{n!} x^n$

$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)a^{n-2}b^2}{2!} + \frac{n(n-1)(n-2)a^{n-3}b^3}{3!} + \cdots$

Power series (convergent for range of x shown):

$$(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots \quad (|x| < 1)$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots \quad (|x| < \pi/2)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad (\text{all } x)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \quad (\text{all } x)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad (|x| < 1)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \quad (\text{all } x)$$